

Article

Substandard and Falsified Antibiotics Seized in Belgium: Quality Control Analysis Reveals High Prevalence of WHO Watch List Molecules and Bioavailability Non-Compliance

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Abstract

Background: Antimicrobial resistance (AMR) poses a critical global public health challenge requiring comprehensive intervention strategies, including robust antibiotic stewardship programs. The European Commission's 2017 One Health Action Plan against AMR established guidelines based on WHO's AWARe classification system, which categorizes antibiotics into Access, Watch, and Reserve groups to promote prudent use. However, the proliferation of substandard and falsified (SF) medical products increasingly undermines these stewardship efforts, with European regulatory agencies, including Belgium's Federal Agency for Medicines and Health Products (FAMHP), reporting rising seizures of SF antibiotics. **Objective:** To assess the pharmaceutical quality of confiscated antibiotic samples and evaluate their potential contribution to AMR development. **Methods:** We conducted comprehensive pharmaceutical quality control testing on 40 SF antibiotic samples seized by the FAMHP between early 2024 and late 2025. **Results:** The analysis revealed significant deficiencies: 35% of samples contained antibiotics listed on international watch lists, while nearly 43% failed quality control testing. Dissolution defects represented the predominant failure mode, accounting for 29% of all samples tested. These defects can severely compromise drug bioavailability, clinical efficiency, and expose bacterial populations to sub-lethal concentrations of active pharmaceutical ingredients—a recognized driver of resistance development. **Conclusions:** Many seized samples appeared to be unregistered/unlicensed medicines that, while prohibited in the EU, may circulate legally elsewhere. This transnational dimension highlights how substandard products threaten global AMR control initiatives beyond individual patient safety. These findings underscore the urgent need to raise more awareness within the EU and for enhanced pharmaceutical quality assurance systems and much more international regulatory cooperation, particularly in low- and middle-income countries where such products circulate more readily.



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1. Introduction

Antimicrobial resistance (AMR) is a multiscale global public health threat that was already responsible for almost five million deaths in 2021 and, if not addressed urgently,

could lead to an estimated death toll of 39 million in the next 25 years [1]. From 1990 to 2021, deaths from AMR increased by over 80% for adults 70 years and older. With a significant decline in the number of new antibiotics released onto the market and a simultaneous increase in resistance to the ones currently in use, it is imperative that the emergence and spread of AMR be monitored, and efforts are made to curb the issue. One of the approaches used to tackle the problem of AMR is antimicrobial stewardship (AMS) [2,3]. The primary objective of AMS is to achieve the best possible clinical outcomes while minimizing adverse effects for patients and minimizing resistance development. In essence, AMS encompasses a coherent set of coordinated actions designed to promote the responsible and judicious use of antimicrobials across all healthcare settings. Moreover, the principles underlying AMS are not limited to human medicine but extend to veterinary and agricultural applications, emphasizing the One Health approach that recognizes the interconnected nature of human, animal, and environmental health [4]. This holistic perspective acknowledges that antimicrobial use in any sector can influence resistance patterns across all domains. In recognition of these interconnections, the European Commission adopted the European Union One Health Action Plan against AMR in June 2017 [5]. As part of implementing this comprehensive action plan, the Commission developed EU Guidelines on the prudent use of antimicrobials in human health, specifically targeting the reduction of inappropriate antimicrobial use while promoting responsible prescribing practices among healthcare providers. The guidelines are based on the World Health Organization's (WHO) AWaRe classification, which divides antibiotics into Access, Watch, and Reserve groups. 'Access' antibiotics are mostly first- and second-line therapies that offer the best therapeutic value, while minimising the potential for antimicrobial resistance (AMR) [6]. 'Watch' antibiotics have a broad spectrum of activity and stewardship efforts that should limit empiric use of these antibiotics to severe infections or infections that are more likely to be resistant to 'Access' antibiotics. 'Reserve' antibiotics include antibiotics of last resort and should be saved for treatment of multidrug-resistant organisms [7]. The European Centre for Disease Prevention and Control (ECDC) and other bodies monitor antibiotic consumption in the EU/EEA using this framework to evaluate stewardship policies and progress toward targets [8].

Complicating these stewardship efforts is the growing problem of substandard and falsified (SF) medical products [9–12]. The WHO has established a uniform definition encompassing three mutually exclusive categories: substandard medical products, unregistered/unlicensed medical products, and falsified medical products [13]. Substandard products, also known as "out of specification" products, originate from legitimate manufacturers but suffer from quality deficiencies that render them unsuitable for use. These products typically enter markets through fraudulent practices, including theft from legitimate supply chains. Unregistered or unlicensed medicines represent products that lack approval from national regulatory authorities in the markets where they are sold, while falsified products deliberately misrepresent their identity, composition, or origin through fraudulent means. The prevalence of SF antibiotics varies across regions: In Asia, the occurrence of SF antibiotics remains below 40%, while South America shows a prevalence of SF antibiotics under 20%, despite a higher AMR burden [11]. Canada reports low levels of both SF antibiotics and AMR, whereas the USA has around 30% of antibiotics classified as SF, with AMR close to 20%. Mexico shows the highest occurrence with 43% of antibiotics being SF. In Western Europe, the presence of SF antibiotics is very low, reflecting strong regulatory oversight. Often, these SF medicinal products are offered through illegal or rogue internet pharmacies that can supply European patients by post or courier [14,15]. These online vendors are neither authorised to operate in the EU nor do they adhere to national practices and guidelines—for example, they offer antibiotics for sale without a prescription.

To prevent this, all online pharmacies in the EU are currently required to display a logo, which acts as a direct link, enabling the checking of the legal status of the pharmacy via the Member State's official pharmacy regulator [16]. Moreover, Belgian law also mandates the collection of prescription antibiotics from a physical, brick-and-mortar pharmacy.

The therapeutic risks posed by substandard and falsified (SF) antibiotics are profound and multifaceted. These products may contain no active pharmaceutical ingredients whatsoever, incorporate incorrect ingredients, deliver improper dosages, demonstrate reduced bioavailability, or harbor dangerous biological or chemical contaminants [17–24]. Beyond these immediate safety concerns, mounting evidence demonstrates a critical link between SF medicines with inadequate active components and the emergence of AMR [9,11,25]. In this study, we conducted comprehensive chemical and biological pharmaceutical quality control testing on 40 SF antibiotic samples confiscated by Belgian regulatory agencies between early 2024 and the end of 2025. Our findings reveal an alarming pattern of deficiencies: 35% of samples contained molecules classified on the WHO's Watch list, while nearly 43% exhibited severe quality control failures. These failures predominantly involved dosage irregularities or dissolution defects, both of which compromise bioavailability and expose bacterial populations to sub-lethal concentrations of active pharmaceutical ingredients—a key driver of AMR development. The implications of these findings extend far beyond individual patient harm. Many of the analyzed antibiotics are strongly suspected to be unregistered/unlicensed medicines or falsifications thereof, which, although prohibited within the European Union, may still circulate legally or illicitly in other regions of the world [15].

2. Results

During the sampling period (2024–2025), thousands of courier and postal packages were seized and analysed by the Belgian Federal Agency for Medicines and Health Products (FAMHP) during routine control operations (e.g., parcels from various courier services, luggage inspections, . . .). From this pool of packages, hundreds of packages were found to contain antibiotics. Only samples intended for oral or injectable use, as they represent the vast majority of the samples encountered by the FAMPH, were considered. Moreover, only those samples that were present in sufficient quantity (more than 20 single dosage units) and not involved in an ongoing investigation were included in this study. A total of 40 samples were retained. An overview of these samples ($n = 40$), along with label information and images of their primary packaging, can be found in Supplemental Table S1. Almost all samples listed essential details: the lot number, expiration date, manufacturer, and indicated that the product requires a medical prescription. However, two samples (numbers 34 and 40) were missing both the manufacturer's information and the prescription requirement notice.

2.1. Types of Antibiotics Found in the Sample Set

Antibiotics can be naturally produced by organisms, semi-synthetically derived, or fully synthetically synthesized. This highly diverse group of molecules is classified into different classes based on chemical structure. However, antibiotics within each class often exhibit distinct pharmacological properties and antimicrobial spectra. As shown in Figure 1, ten different antibiotic classes were identified in our sample set. Although the majority of SF antibiotics belonged to the β -lactams (38%), tetracyclines and macrolides were also well represented at 20% and 18%, respectively. Additionally, antibiotics from the fluoroquinolone, nitroimidazole, rifamycin, cephalosporin, ethylenediamine derivative, pyridine derivative, and macrolactam classes were identified. The most frequently encountered SF antibiotics were amoxicillin (with or without clavulanic acid), azithromycin, doxycycline,

and rifaximin. The majority of detected active pharmaceutical ingredients (APIs) exhibited broad-spectrum antimicrobial activity, while five antibiotics were characterized by narrow-spectrum activity. These are cloxacillin, primarily used to treat infections caused by Gram-positive *Staphylococcus* spp.; phenoxymethylpenicillin targeting Gram-positive cocci (e.g., *Streptococcus* spp.); metronidazole targeting anaerobic bacteria and protozoa; and ethambutol and isoniazid are used against *Mycobacterium* species [26].

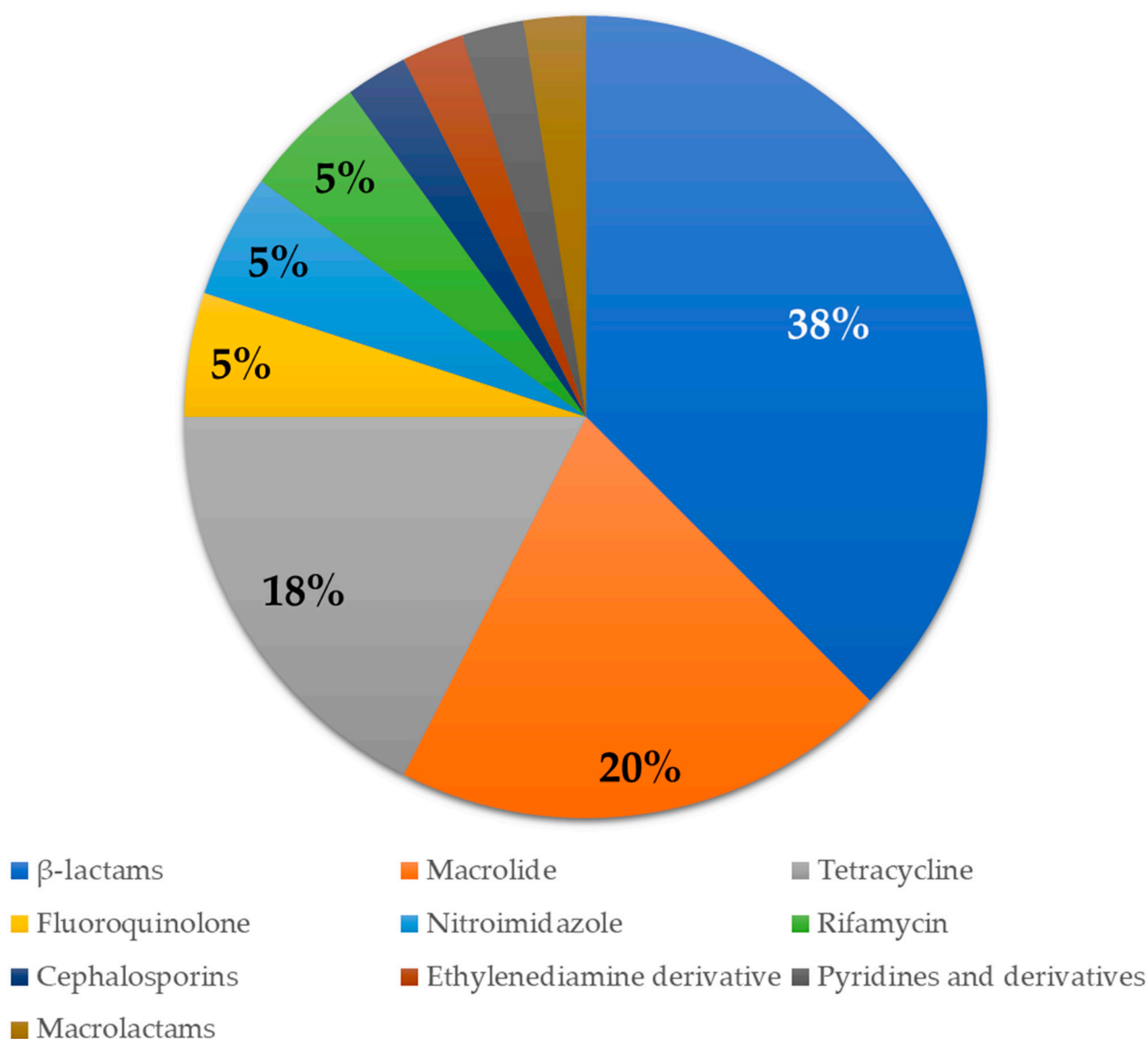


Figure 1. Abundance of the different antibiotic classes represented in the SF medicines. Only abundance percentages $\geq 5\%$ are shown. Abundances below 5% represent only one sample from the sample set.

Strikingly, six identified antibiotics—azithromycin, rifaximin, ceftriaxone, ciprofloxacin, levofloxacin, and rifampicin—are included on the WHO Watch List [6], collectively accounting for 35% of the sample set.

2.2. Chemical Quality Control

The samples were initially checked to confirm they contained the APIs listed and screened for the presence of unlisted APIs. The amount of each API was assessed using liquid chromatography coupled with a PDA detector, employing methodologies

specified in the United States Pharmacopoeia (USP) for finished products containing the API(s) or methodologies that were reported and validated in the literature (see supplemental Table S2). Only for ceftriaxone, a novel methodology was developed specifically for this study. For each testing method, a mini validation was performed to demonstrate fit for purpose, as is done routinely in our Official Medicines Control Laboratory (OMCL). The analysis also applied the requested quality criteria for chromatographic peaks as specified in both the European Pharmacopoeia [27] and USP [28]. As shown in Table 1, seven of the 40 samples, equivalent to 18% of the samples, failed to meet the requirements put forward by the USP. In the case of the presence of the API amoxicillin, amoxicillin and clavulanic acid, phenoxymethyl penicillin, azithromycin, and doxycycline, these samples contained either less than 90% or more than 120% of the declared amount. Notably, sample 33 contained no detectable amounts of clavulanic acid, although the label listed the presence of 125 mg per tablet. In the case of levofloxacin, tinidazole, ciprofloxacin, metronidazole, rifampicin, and isoniazid, the samples should contain between 90 and 110% of the labelled API, while the ethambutol samples should contain between 95 and 10%, and the ceftriaxone for injection purposes should contain between 90 and 115% of the labelled amount.

While 82% of the samples contained the correct amount of API, this alone doesn't guarantee the drug's effectiveness. Therefore, the dissolution assay—which measures how quickly and completely a tablet or capsule releases its API—is essential to confirm proper release and bioavailability of the APIs. The dissolution assays were performed on 35 of the 40 samples, as 5 samples lacked a monograph describing an assay. As shown in Table 1, approximately 29% of the tested samples (10/35) did not meet the dissolution requirements outlined in the USP (see supplemental Table S3). Among these failing samples, six released less than 60% of their API during the test, suggesting the medicine might not dissolve properly in the body. Only for 3 samples could this non-compliance be attributed to dosage irregularities.

Next, the samples were tested for residual solvents. As shown in Table 1, both rifaximin samples (samples 29 and 30, representing 5% of the sample set) exceeded the USP limits for dichloromethane and 2-propanol, solvents that can be used in the synthesis of this antibiotic. Dichloromethane is classified as a class 2 solvent, which should be limited due to its toxicity, with a maximum allowed level of 600 parts per million (ppm) per tablet [28]. Both samples contained at least twice this amount. Additionally, higher-than-tolerated levels were also found for the class 3 solvent, 2-propanol, with quantities exceeding at least five times the limit of 5000 ppm. The quantifiable residual solvent levels detected in all samples are provided in Supplemental Table S4.

Table 1. Overview listing the results of the chemical and biological quality control analysis.

n°	API Found	Anti-Biotic Class	Amount Claimed (mg)	% Labeled Amount Found (Mean, MU ^a)	Dissolution % (Mean, MU ^a)	Residual Solvents Surpassing the Tolerated Concentration Limit (ppm)	CFU/g or mL	TAMC Identified Organism	CFU/g or mL	TYMC Identified Organism
1	Doxycycline	Tetracycline	100	93.3 ± 1.6	82.6 ± 5.0	-	n.d.	n.d.	n.d.	n.d.
2	Doxycycline	Tetracycline	100	79.3 ± 1.3	43.2 ± 6.4	-	≈200	Fungal species grew on the medium	≈300	<i>Aspergillus sydowii</i> and <i>Aspergillus rhizopodus</i>
3	Doxycycline	Tetracycline	100	92.7 ± 4.5	68.9 ± 8.4	-	≈1000	Fungal species grew on the medium	≈1600	<i>Aspergillus hortae</i> , <i>Aspergillus sydowii</i> , <i>Aspergillus rhizopodus</i> , <i>Aspergillus chevalieri</i> and <i>Paecilomyces formosus</i>
4	Amoxicillin	β-lactam	500	91.5 ± 0.7	75.5 ± 2.8	-	n.d.	n.d.	n.d.	n.d.
5	Amoxicillin	β-lactam	500	88.6 ± 1.7	45.0 ± 6.8	-	n.d.	n.d.	n.d.	n.d.
6	Azithromycin	Macrolide	500	99.5 ± 0.6	98.3 ± 6.6	-	n.d.	n.d.	n.d.	n.d.
7	Azithromycin	Macrolide	500	98.1 ± 0.1	94.05 ± 6.3	-	n.d.	n.d.	n.d.	n.d.
8	Azithromycin	Macrolide	500	102.6 ± 0.6	104.8 ± 1.8	-	n.d.	n.d.	n.d.	n.d.
9	Amoxicillin	β-lactam	250	101.0 ± 1.3	86.9 ± 6.1	-	n.d.	n.d.	n.d.	n.d.
9	Clavulanic acid	β-lactamase inhibitor	125	105.0 ± 1.7	96.8 ± 5.9	-	n.d.	n.d.	n.d.	n.d.
10	Ciprofloxacin	Fluoroquinolone	500	91.6 ± 1.2	n.a.	-	n.d.	n.d.	n.d.	n.d.
10	Tinidazole	Nitroimidazole	600	109.1 ± 0.2	n.a.	-	n.d.	n.d.	n.d.	n.d.
11	Phenoxyethyl penicillin	β-lactam	250	80.8 ± 0.2	81.3 ± 1.5	-	n.d.	n.d.	n.d.	n.d.
12	Doxycycline	Tetracycline	100	94.8 ± 2.9	88.5 ± 22.7 *	-	n.d.	n.d.	n.d.	n.d.
13	Azithromycin	Macrolide	250	91.1 ± 0.8	58.2 ± 2.4	-	n.d.	n.d.	n.d.	n.d.
14	Amoxicillin	β-lactam	1000	86.8 ± 0.5	54.2 ± 5.0	-	n.d.	n.d.	n.d.	n.d.
15	Amoxicillin	β-lactam	875	89.2 ± 2.4	89.1 ± 2.5	-	≈60	<i>Streptomyces calvus</i> and <i>Kocuria rhizophila</i>	n.d.	n.d.
16	Clavulanic acid	β-lactamase inhibitor	125	101.3 ± 1.0	78.6 ± 9.9	-	n.d.	n.d.	n.d.	n.d.
17	Ceftriaxone	Cephalosporin	1000	76.7 ± 1.8	n.a.	-	n.d.	n.d.	n.d.	n.d.
18	Amoxicillin	β-lactam	500	81.5 ± 0.5	86.5 ± 6.8	-	n.d.	n.d.	n.d.	n.d.
19	Azithromycin	Macrolide	500	93.4 ± 0.7	91.6 ± 6.9	-	≈20	<i>Kocuria rhizophila</i>	n.d.	n.d.
20	Amoxicillin	β-lactam	500	89.0 ± 1.1	72.2 ± 2.1	-	n.d.	n.d.	n.d.	n.d.
21	Doxycycline	tetracycline	100	99.4 ± 4.1	87.2 ± 1.2	-	n.d.	n.d.	n.d.	n.d.
22	Metronidazole	Nitroimidazole	400	97.8 ± 0.4	104.5 ± 1.2	-	n.d.	n.d.	n.d.	n.d.
23	Doxycycline	tetracycline	100	103.9 ± 3.5	98.0 ± 3.6	-	n.d.	n.d.	n.d.	n.d.
24	Doxycycline	tetracycline	100	102.4 ± 2.9	93.6 ± 3.5	-	n.d.	n.d.	n.d.	n.d.
25	Amoxicillin	β-lactam	500	98.2 ± 1.0	61.9 ± 10.9	-	≈20	<i>Paenibacillus</i> spp.	n.d.	n.d.
26	Amoxicillin	β-lactam	500	99.5 ± 0.3	83.3 ± 10.1	-	n.d.	n.d.	n.d.	n.d.
27	Clavulanic acid	β-lactamase inhibitor	125	101.9 ± 2.4	84.4 ± 5.5	-	n.d.	n.d.	n.d.	n.d.
28	Amoxicillin	β-lactam	500	136.5 ± 1.6	98.8 ± 10.8	-	n.d.	n.d.	n.d.	n.d.
29	Clavulanic acid	β-lactamase inhibitor	125	134.2 ± 2.2	97.5 ± 4.3	-	n.d.	n.d.	n.d.	n.d.
30	Amoxicillin	β-lactam	250	102.3 ± 1.6	n.a.	-	n.d.	n.d.	n.d.	n.d.
31	Cloxacillin	β-lactam	250	99.6 ± 1.0	n.a.	-	n.d.	n.d.	n.d.	n.d.
32	Levofloxacin	Fluoroquinolone	500	101.9 ± 0.9	105.8 ± 0.4	-	n.d.	n.d.	n.d.	n.d.
33	Rifaximin	Rifamycin	200	102.8 ± 0.2	n.a.	Dichloro-methane and 2-propanol ^b	n.d.	n.d.	≈40	<i>Cladosporium allcinum</i>
34	Rifaximin	Rifamycin	550	106.3 ± 0.3	n.a.	Dichloro-methane and 2-propanol ^b	n.d.	n.d.	≈100	<i>Cladosporium inversicolor</i> and <i>Cladosporium allcinum</i>
35	Azithromycin	Macrolide	250	98.9	96.1 ± 2.7	-	n.d.	n.d.	n.d.	n.d.
36	Azithromycin	Macrolide	250	98.5	41.2 ± 6.6	-	n.d.	n.d.	n.d.	n.d.
37	Amoxicillin	β-lactam	500	106.8 ± 4.0	82.0 ± 1.3	-	n.d.	n.d.	n.d.	n.d.
38	Clavulanic acid	β-lactamase inhibitor	125	n.d.	n.d.	-	n.d.	n.d.	n.d.	n.d.
39	Amoxicillin	β-lactam	250	117.3 ± 4.5	81.2 ± 2.4	-	n.d.	n.d.	n.d.	n.d.

Table 1. Cont.

n°	API Found	Anti-Biotic Class	Amount Claimed (mg)	% Labeled Amount Found (Mean, MU ^a)	Dissolution % (Mean, MU ^a)	Residual Solvents Surpassing the Tolerated Concentration Limit (ppm)	CFU/g or mL	TAMC Identified Organism	CFU/g or mL	TYMC Identified Organism
35	Ethambutol	ethylenediamine derivative	800	100.1 ± 7.1	102.3 ± 1.1	-	≈1000	<i>Bacillus badius</i> and <i>Paenibacillus</i> spp.	≈300	<i>Cladosporium allicinum</i> and <i>Cladosporium</i> spp.
36	Rifampicin	Macrolactam	600	92.0 ± 1.0	75.8 ± 4.0	-	n.d.	n.d.	n.d.	n.d.
	Isoniazid	Pyridine and derivatives	300	90.7 ± 0.8	79.9 ± 5.4	-				
37	Amoxicillin	β-lactam	500	92.6 ± 6.5	86.0 ± 0.4	-	n.d.	n.d.	n.d.	n.d.
38	Clavulanic acid	β-lactamase inhibitor	125	110.5 ± 1.3	101.6 ± 0.3	-	n.d.	n.d.	n.d.	n.d.
39	Azithromycin	Macrolide	500	101.6 ± 0.6	102.2 ± 0.4	-	n.d.	n.d.	n.d.	n.d.
40	Metronidazole	Nitroimidazole	400	97.7 ± 0.5	99.0 ± 0.8	-	≈60	<i>Kocuria rhizophila</i>	n.d.	n.d.
	Amoxicillin	β-lactam	500	94.4 ± 2.3	74.6 ± 7.2	-	n.d.	n.d.	n.d.	n.d.

^a The measurement uncertainty (MU) is expressed as a confidence interval using the standard deviation of the generated quantification results. ^b The limits for dichloromethane correspond to 600 ppm, while the one for 2-propanol is set at 5000 ppm. For details of the amount of residual solvent found, see Supplemental Table S4. Abbreviations: CFU: colony forming units; MU measurement uncertainty; n.a.: not applicable; n.d.: none detected; ppm: parts per million; TAMC: total aerobic count; TYMC: total yeast and mould count. A star (*) indicates an exceptionally large measurement MU, suggesting large differences between capsules.

2.3. Biological Quality Control

Beyond chemical quality control, non-sterile pharmaceutical products must also be checked for biological contamination, including bacteria and fungi. Some microbial presence is tolerated, but only at levels that ensure patient safety. When contamination exceeds the limits set by pharmacopoeias, there's a risk of adverse health effects for patients. According to the USP, the acceptance criteria for finished product type tablets and capsules correspond to 103 colony-forming units (CFU)/g in the total aerobic microbial count (TAMC) test (mainly bacteria) and 102 CFU/g in the total yeast and mould count (TYMC) test [28]. The results, presented in Table 1, show that the majority of the samples were compliant and not contaminated with known pathogens, except for 3 samples, accounting for 7.5% of the sample set. Two of these samples (Samples 2 and 3) were severely contaminated with fungi from the *Aspergillus* genus. One of the *Aspergillus* species found in sample 3 corresponded to *Aspergillus sydowii*, a known opportunistic pathogen that can cause oral aspergillosis in immunocompromised patients [29]. Moreover, another opportunistic pathogen, *Paecilomyces formosus*, was found in this sample, although it rarely results in clinically relevant illness [30].

3. Discussion

Ten different antibiotic classes were detected in our sample set encompassing forty different seizures, clearly illustrating the wide variety of SF antibiotics circulating through illegal channels. However, the limited sample size ($n = 40$) and specific selection criteria restrict the generalizability of these findings. Only samples intended for oral or injectable use, available in sufficient quantities (>20 dosage units), and not involved in ongoing investigations were included. Additionally, the predominance of unregistered/unlicensed medicines introduces selection bias, as falsifications of registered and licensed medicines were excluded. These limitations should be considered when interpreting the results. Nevertheless, the dominance of β -lactams (38%), tetracyclines (18%), and macrolides (20%) matches their common therapeutic use. Compared to a previous study analysing samples seized between 2015 and 2019 [17] (see Supplemental Figure S1), there is noticeably more diversity in the type and amount encountered. Furthermore, the proportion of samples containing either a macrolide or a tetracycline has increased, from 14% to 20% and from 7% to 18%, respectively. Notably, the presence of six antibiotics on the WHO Watch List—accounting for more than one-third of the sample set—raises serious concerns. The use of these agents (e.g., azithromycin, rifaximin, and fluoroquinolones) should be limited due to their high potential to promote antimicrobial resistance (AMR) when misused or of substandard quality.

Additionally, the self-medication with certain antibiotics is certainly not without risks, as it can lead to severe allergic reactions and antibiotic-associated diarrhoea. Moreover, some antibiotics, especially fluoroquinolones, are also well known for their disabling and potentially permanent side effects (see Table 2). Consequently, in 2019, the European Commission, based on studies performed by the European Medicines Agency (EMA), imposed restrictions on the use of fluoroquinolones across the EU [31]. Moreover, other severe side effects, beyond the standard antibiotic risks such as allergic reactions and antibiotic-associated diarrhoea, were observed with anti-tuberculosis (TBC) antibiotics (rifampicin, isoniazid, and ethambutol), azithromycin, and doxycycline (see Table 2).

Table 2. Overview of the health risks associated with the use of certain APIs, beyond the standard antibiotic risks such as allergic reactions and antibiotic-associated diarrhoea.

API	Sample Number	Known Risks
Ciprofloxacin	10	Since 2019, the European Medicines Agency (EMA) has proposed restricting the use of fluoroquinolone antibiotics due to severe, potentially disabling, and potentially permanent side effects [31]. These serious side effects include inflamed or torn tendons, muscle pain or weakness, joint pain or swelling, walking difficulty, feeling pins and needles, burning pain, tiredness, depression, problems with memory, sleep, vision, and hearing, and altered taste and smell.
Levofloxacin	28	
Rifampicin and isoniazid	36	Rifampicin is associated with rare instances of clinically apparent liver injury accompanied by symptoms and jaundice, which can be severe and even fatal [32]. The combination of rifampicin with isoniazid, another known hepatotoxic agent, is more likely to cause injury than the individual drugs would suggest.
Ethambutol	35	Ethambutol use may lead to permanent vision loss by inducing a dose- and duration-dependent optic neuropathy [33]. Studies have shown that the incidence of ethambutol-induced ocular neuropathy corresponds to 1 to 2%. Susceptibility to toxicity increases in patients suffering from alcohol addiction, tobacco addiction, diabetes, hypertension, renal diseases, and liver diseases.
Azithromycin	6–8, 13, 18, 31, 32 and 38	Azithromycin use may lead to a potentially fatal irregular heart rhythm [34,35]. Patients at particular risk for developing this condition include those with known risk factors such as existing QT interval prolongation, low blood levels of potassium or magnesium, a slower-than-normal heart rate, or use of certain drugs used to treat abnormal heart rhythms or arrhythmias.
Doxycycline	1–3, 12, 20, 22 and 23	The usage of Doxycycline can result in serious skin reactions, including exfoliative dermatitis, erythema multiforme, Stevens–Johnson syndrome (SJS), and toxic epidermal necrolysis [36]. Additionally, although very rare, the use of this drug might result in reversible intracranial hypertension [37].

Rifampicin, when combined with isoniazid, is known to cause hepatotoxicity [32], while ethambutol, the other anti-TBC antibiotic in the sample set, can lead to vision loss due to optic neuropathy [33]. Azithromycin, on the other hand, although rare, can induce irregular heart rhythms [34,35], while the use of doxycycline can result in serious skin reactions [36] and intracranial hypertension [37]. Moreover, self-medication with these prescription medicines could cause undesired side effects through interactions with other medications the patient is taking.

Chemical and biological quality controls performed on these SF antibiotics reveal significant public health risks: 18% of samples contained incorrect amounts of the declared active pharmaceutical ingredient (API), while 29% failed dissolution tests. Although most products contained the correct API, deviations from pharmacopoeial dissolution specifications can severely impair oral bioavailability and consequently can result in sub-therapeutic peak plasma concentrations, which may compromise the drug's ability to reach the minimum inhibitory concentration at the site of infection, reducing bactericidal activity and clinical effectiveness. Specifically, the dissolution failure of Azithromycin (samples 13 and 32) is particularly concerning, as this API is classified as a BCS II (Biopharmaceutics Classification System) drug, characterized by low solubility. As a result, the observed dissolution issues will very likely directly affect its bioavailability. Moreover, the complete absence of clavulanic acid in one sample labelled as amoxicillin-clavulanate is also quite worrisome. Subtherapeutic dosing and inadequate API release not only cause therapeutic

failure but also drive the selection and spread of resistant bacterial strains, thereby exacerbating the global AMR crisis [9,11,25]. Additionally, two rifaximin-containing samples exceeded maximum permitted levels for the Class 2 solvent dichloromethane (DCM) and the Class 3 solvent 2-propanol. However, the detected quantities—up to 1.6 mg DCM and 28.5 mg 2-propanol—are unlikely to cause acute intoxication in adults. For DCM, at least 0.5–5 mL/kg body weight is required to produce severe effects such as central nervous system depression, metabolic acidosis, and carboxyhemoglobinemia [38–40]. For 2-propanol, quantities exceeding 2–4 mL per kg were reported to produce acute toxic effects in adults, ranging from gastrointestinal irritation to inebriation, haemorrhagic gastritis, and even coma and death [41]. Nevertheless, rifaximin may be used over extended periods, even by children [42,43]; consequently, potential chronic exposure effects cannot be excluded. Microbiological analysis revealed severe fungal contamination in three samples. One sample was contaminated with the opportunistic pathogens *Aspergillus sydowii* and *Paecilomyces formosus* [29,30]. Although severe human infections are rare, the presence of these fungi is unacceptable. It represents a clear safety hazard, likely resulting from inadequate environmental controls during production or deficient packaging and storage conditions. Collectively, these findings expose multidimensional quality deficiencies of unregistered antibiotic samples or falsifications thereof.

Alarming, multiple previous studies have demonstrated that many of these unregistered antibiotics enter the legal supply chain in low and middle-income countries (LMICs), which already have a high burden of AMR [15,24]. It makes sense that high demand, along with limited resources, weak regulatory control systems, and poor importation standards, contribute to the persistence of these issues in those areas. In contrast, the EU's regulatory control systems are much stricter, particularly since the implementation of the Falsified Medicines Directive to improve protection of the legal medicines supply chain and curtail the occurrence of SF medicines [44]. Mandatory measures include stricter record-keeping by wholesale distributors, tougher inspections of pharmaceutical producers, and an EU-wide quality mark to identify legitimate online pharmacies. Moreover, additional safety features, such as a unique identifier on each package encoded in a two-dimensional data matrix, are required. This code, printed on each unit-of-sale package, must be uploaded to the EU Hub of the European Medicines Verification System (EMVS), a central database for verifying the authenticity of prescription medicines across the EU. As no such data matrix was found on the seized packages, these samples likely originated from the illegal supply chain. This conclusion is further supported by the discovery of antibiotics in postal packages, which is particularly significant given that Belgian law requires a prescription for purchasing antibiotics and mandates collection from a physical, brick-and-mortar pharmacy rather than direct shipment to consumers. Worryingly, the occurrence of these products in the EU, where antibiotic treatment is generally considered affordable, except for tuberculosis, is rather unexpected. The explanation for this behavioural phenomenon is likely multifaceted and may be driven by financial considerations, as some international online pharmacies operating illegally outside the EU sell blister packs of oral antibiotic treatments at extremely low prices. Moreover, 2023 and 2024 were marked by shortages of different medicinal products in the EU, including antibiotics [45]. It is possible that people facing unmet medical needs or medicine demands were tempted by the ease of online purchasing. Additionally, Belgium's stricter antibiotic prescription policies—part of antimicrobial stewardship (AMS) measures—combined with patients' belief that they need antibiotics, may have led to an influx of antibiotics obtained from alternative sources, including online. Moreover, the FAMPH also noticed that in recent years, there has been an increase in packages sent from non-EU family members to Belgian family members containing not only herbs and clothing, but also antibiotics. Taken together, this indicates that restricting access to an-

tibiotics in the (Belgian) legal supply chain creates higher demand, which is exploited by malicious actors. It stands to reason that more coordinated global efforts are needed to raise awareness about the dangers of irresponsible antibiotic use and their online purchasing [13,46–51]. Moreover, coordinated global initiatives, including also the big production countries, are required to curtail irresponsible antibiotic manufacturing. These should include strengthening regulatory capacity and post-marketing surveillance in all countries; harmonizing quality standards and enforcement mechanisms internationally; investing in quality control infrastructure and human resources; implementing track-and-trace systems to secure pharmaceutical supply chains; and integrating pharmaceutical quality assurance into national AMR action plans [49–51]. This is to ensure that existing AMS programs are not compromised or worsen the AMR burden. Certainly, bearing in mind that the latest WHO data show that one in six bacterial infections worldwide now exhibits resistance to common antibiotics, underscoring the urgency of such actions [52].

4. Materials and Methods

4.1. Standards, Solvents and Reagents

Mass spectrometry (MS)-grade methanol (purity > 99.9%), acetonitrile (purity > 99.9%), and water were purchased from Thermo Fisher Scientific (Waltham, MA, USA), while MS-grade formic acid (purity > 99%) was purchased from Biosolve (Valkenswaard, the Netherlands). The reference standards used for confirmation of the identification and quantification assays were obtained from Merck (Darmstadt, Germany). Their percentage purity and the solvents used to prepare a 0.2 mg/mL stock solution for MS confirmation are listed in Supplemental Table S5.

The solvents (methanol and acetonitrile) used for the quantification of the active pharmaceutical ingredient (API) were high-performance liquid chromatography (HPLC)-grade and purchased from Biosolve (Valkenswaard, the Netherlands). Water was generated using a Milli-Q Gradient A10 system (Millipore, Billerica, MA, USA). All other chemicals used in the quantification and dissolution assay, including hydrogen chloride, phosphoric acid, sodium hydroxide, KH_2PO_4 , Na_2HPO_4 , NaH_2PO_4 , ammonium acetate, disodium ethylenediaminetetraacetate (Na_2EDTA), triethylamine, L-isoleucine, CuSO_4 , glacial acetic acid and Dimethyl Sulfoxide (DMSO), used as a solvent for the residual solvent analysis of the different samples were purchased from Merck (Darmstadt, Germany). HPLC-grade 2-propanol, acetone, dichloromethane, and pesticide-S-grade ethyl acetate, used to quantify the detected residual solvents, were also purchased from Biosolve (Valkenswaard, the Netherlands). Chloroform was purchased from Thermo Scientific (Waltham, MA, USA).

4.2. Buffers and Media for Bioburden Testing

Sabouraud dextrose agar with neutralisers (lecithin, tween and histidine), tryptocasein-soy agar with neutralisers (lecithin, tween, histidine and thiosulfate), violet red bile glucose agar, and buffered sodium chloride peptone with neutralisers (lecithin, tween, histidine and thiosulfate) were purchased from Merck (Darmstadt, Germany). Sterile analytical filter units 0.45 μm Nalgene™, used for the membrane filtration during the bioburden testing, were purchased from Thermo-Fisher Scientific (Rochester, NY, USA).

4.3. Sample Set

Samples were seized by the Belgian Federal Agency for Medicine and Health Products (FAMHP) and the samples that met the predefined criteria (see Results) were sent to our Official Medicines Control Laboratory (OMCL). Pictures were taken upon arrival, and the information available on the packaging or blister was recorded (see Supplementary Table S1).

All, except one, samples were in the form of tablets or capsules. One sample was for injection purposes. All samples were analyzed before the expiration date.

4.4. Screening for Active Pharmaceutical Ingredients

All samples were screened for synthetic drugs/medicines using LC-MS² as previously described [53]. In addition to the solubilization in methanol, as is done routinely, the samples labeled to contain a certain antibiotic were also solubilized in their most optimal solvent for the specific API (see Supplemental Table S5). The LC-MSⁿ screening was performed on a Dionex UltiMate 3000 RSLC system (Thermo Scientific, Waltham, MA, USA) connected to an amaZonTM speed ETD mass spectrometer (Bruker Daltonics, Billerica, MA, USA). Data were analyzed with Compass Data Analysis 4.2 (Bruker Daltonics, Billerica, MA, USA) and compared against in-house libraries containing ~5000 MS/MS² spectra. Compounds were identified when: retention time shift ≤ 0.5 min, precursor ion m/z matched the in-house library (± 0.3 Da), and MS² spectra showed $\geq 85\%$ match with reference spectra.

4.5. Quantification of Active Pharmaceutical Ingredients

The API content in samples was determined by liquid chromatography using either a Waters Alliance HPLC or Acquity UPLCTM system (Waters Corp., Milford, MA, USA) coupled to a sample manager with flow-through needle, column heater, and photo-diode array (PDA) detector connected to a Waters Empower 3.8.1 data station. Chromatographic separation conditions were selected based on the APIs present in each sample. Details of the methodologies used, predominantly based on United States Pharmacopoeia (USP) [28] or previously published methods [17,54–57], are provided in Supplementary Table S2. For the quantification of ceftriaxone, preference was given to the use of a simple and fast ultra-high-performance liquid chromatography (UHPLC) methodology that was previously developed in our laboratory. Briefly, the chromatographic separation was performed at 25 °C on an AcquityTM UPLC CSH Column (100 $\times$ 2.1 mm, 1.7 μ m particle size) (Waters, Milford, MA, USA) with a mobile phase consisting of 0.2% formic acid in water (A) and acetonitrile (B) and flow rate of 0.3 mL/min. An isocratic step at 5% B was kept for one min. prior to a linear gradient of 50% B in 10 min. Ceftriaxone was monitored at a wavelength of 280 nm.

Rather than full validation, we performed fit-for-purpose verification—our standard practice in our medicines control laboratory for quality assessment of legal medicines—demonstrating linearity, accuracy ($\pm 5\%$), and sensitivity across 60–140% of the target concentration. Peak identity for each API was confirmed by comparing retention time and ultraviolet absorbance spectra against a reference standard. The purity angle and purity threshold were evaluated for all samples and found to be below the purity threshold, confirming good chromatographic peak purity. Sample quantification was performed on three separate units (capsules, tablets, or vials). Each preparation ($n = 3$) was injected four times, and results from these 12 injections were used to calculate the mean API content. Measurement uncertainty was estimated as a confidence interval based on the standard deviations obtained from these preparations.

4.6. Dissolution Assay

The dissolution analyses were conducted on a Sotax AT Xtend dissolution device (Sotax, Aesch, Switzerland) equipped with six dissolution vessels (dark glass) and baskets or paddles, corresponding to test apparatus 1 and 2 as specified by the Ph. Eur. and USP [27,28]. The temperature of the dissolution medium was maintained at 37.0 ± 0.5 °C, and the various dissolution settings, depending on the galenic form, labeled API amount, and API combination in each sample, are listed in the Supplemental Data S3. For the

quantification of the API or API mixture, the chromatographic methods described in Section 4.5 were employed.

4.7. Screening for Residual Solvents

Samples were analyzed for residual solvents, which were quantified according to previously described methodology [58]. Briefly, the mass spectrometer was operated in full-scan mode for solvent identification and in selected ion monitoring (SIM) mode for quantification. Calibration dilutions were prepared from stock solutions in dimethyl sulfoxide (DMSO), transferred to vials, and automatically sealed. DMSO served as the blank. For sample analysis, 1 mL of DMSO was added to the tablets. Samples were prepared and injected in duplicate. A one-point calibration was performed within the linear range established for each detected residual solvent.

4.8. Bioburden Determination

Bioburden testing aims to quantify aerobic microorganisms that may be present in pharmaceutical preparations. These microorganisms are cultured to form visible colonies, with each colony assumed to originate from a single colony-forming unit (CFU). Bioburden testing was performed according to Ph. Eur. guidelines using the membrane filtration method [27].

Bacteria and fungi detected in the samples were isolated on tryptone-casein-soy agar and Sabouraud medium, respectively, prior to the extraction of biological material for subsequent matrix-assisted laser desorption/ionization time-of-flight mass spectrometry (MALDI-TOF) analysis [19,47,52]. The generated data were processed using MALDI Biotyper 3 software (Bruker Daltonics, Bremen, Germany). Only hits with log(score) values ≥ 2.00 were considered high-confidence identifications at the species level. For certain extracts, only hits with log(score) values between 1.70 and 2.00 were reported and thus resulted in an identification at the genus level.

4.9. Use of GenAI for Graphical Abstract

During the preparation of this manuscript, the author(s) used ChatGPT (version 5.2) to generate parts of the graphical abstract.

5. Conclusions

This study provides a comprehensive assessment of the pharmaceutical quality of substandard and falsified (SF) antibiotics seized in Belgium between 2024 and 2025, highlighting their potential to undermine antimicrobial stewardship efforts and exacerbate the global antimicrobial resistance (AMR) crisis. Although the majority of samples contained the declared active pharmaceutical ingredient, nearly 43% failed at least one critical quality parameter, most notably dissolution testing (failure: 29%), which directly affects bioavailability and therapeutic efficacy. Such deficiencies are particularly concerning given that 35% of the analyzed samples contained antibiotics classified within the WHO Watch group, for which inappropriate use or subtherapeutic exposure poses a heightened risk for resistance selection. Additionally, excessive amounts of residual solvents and unacceptable microbial contamination were found in 5% and 7.5% of the samples, respectively. Taken together, these findings illustrate the multifaceted health risks associated with SF antibiotics.

The presence of these products in a highly regulated European setting suggests that illegal online sales, informal cross-border supply, medicine shortages, and patient-driven demand continue to create vulnerabilities that are exploited by illicit actors. Moreover, many of the analyzed products are suspected to be unregistered or unlicensed medicines that may circulate more widely in low- and middle-income countries, where regulatory oversight is often weaker, and the burden of AMR is already substantial. These findings reinforce the

need for stronger international regulatory cooperation, improved surveillance of antibiotic quality, and greater public awareness of the risks associated with purchasing medicines outside the legal supply chain. Safeguarding antibiotic quality must be recognized as a core component of AMR containment strategies to ensure that existing stewardship efforts are not compromised at national or global levels.

Supplementary Materials: The following supporting information can be downloaded at: <https://www.mdpi.com/article/10.3390/antibiotics15020230/s1>, Figure S1: Abundance of the different antibiotic classes represented in the SF medicines analysed in a previous study performed from 2015 to 2019 [17,18]; Table S1: Overview of the information labeled on the blister or package and description of the appearance of the different samples; Table S2: Overview of the different chromatographic methodologies used to quantify the API present in the samples and the diluent used for sample and reference standard preparation for the different quantification assays. Abbreviations: ACN = acetonitrile, MeOH = methanol [17,28,48–51]; Table S3: Overview of the dissolution assays, if available in the USP, Ph. Eur. and BP, used. Abbreviations: n.a. = not applicable. ^a For this sample, the USP monograph for Rifampin, Isoniazid, Pyrazinamide, and Ethambutol Hydrochloride Tablets was used; Table S4: Overview of the residual solvents found in each sample and the amount present, expressed in parts per million (ppm); Table S5: Overview of the listed purity of the reference standards used and the diluent used to generate a stock solution at 0.2 mg/mL, used for LC-MSⁿ confirmation purposes. Abbreviations: MeOH = methanol, v/v = volume per volume.

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